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3 Hours / 70 Marks Seat No. | | | | | | | | |

Instructions : (1) All Questions are *compulsory.*

(2) Illustrate your answers with neat sketches wherever necessary.

(3) Figures to the right indicate full marks.

(4) Assume suitable data, if necessary.

Marks

1. Attempt any FIVE of the following : 10

(a) State the function of flatbed scanner.

(b) Define bit depth of colour.

(c) Write names of two process inks used in gravure printing process.

(d) State two file formats used in printing industry.

(e) Write two characteristics of GCR.

(f) Define color gamut.

(g) State the visible range of human vision.

2. Attempt any THREE of the following : 12

(a) State four characteristics of subtractive colour theory.

(b) State two characteristics each of any two dot shapes.

(c) State four characteristics of pantone colour specification system.

(d) State four specifications of spectrophotometer.

3. Attempt any THREE of the following : 12

(a) Describe metamerism and adaptation.

(b) Explain Moire defect. State its cause and remedy.

(c) Describe additivity failure and proportionality failure.

(d) Explain ink trapping. State its formula.

4. Attempt any THREE of the following : 12

(a) State two characteristics each of physical and digital originals.

(b) State the meaning of following terms – resolution, ppi, dpi, lpi.

(c) State two characteristics each of vector and raster format.

(d) Describe the meaning of device dependent colours and device independence colours.

(e) Write screen angle for planographic offset printing process for CMYK process inks.

5. Attempt any TWO of the following : 12

(a) Explain additive colour theory with schematic diagram.

(b) State three characteristics each of AM and FM screening techniques.

(c) Describe the working principle and construction of optical densitometer.

6. Attempt any TWO of the following : 12

(a) Differentiate between light and colour up to 6 points.

(b) State four operations performed during preflight check. Also state the purpose of preflight.

(c) Calculate the colour difference (Delta E) between two printed samples.

L1 = 22, L2 = 23, a1 = 24, a2 = 25, b1 = 19, b2 = 24.